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Deep Data Analysis for Liver Cancer Prediction Using Machine Learning Approaches

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Abstract-

The abstract uses a hybridized fully convolutional neural network (HFCNN) to synthesize recent developments in liver cancer diagnosis. This technique improves the accuracy of liver segmentation, which is essential for tumor identification, by combining deep learning with 3D PET-CT imaging. Cutting-edge methods such as multi-view learning enhance the efficacy of the HFCNN in identifying cancers in low-contrast CT scans. The innovation consists of a three-tier segmentation strategy and a special ensemble method that uses deep neural networks and mathematical models for high-performance tumor classification to improve liver tumor segmentation. These techniques have shown promise in improving early diagnostic and treatment decision-making for liver cancer, as evidenced by their performance in clinical datasets. The integrated approach provides a useful resource for medical imaging and represents a substantial development in the diagnosis of liver cancer.

INDEX TERMS *Liver cancer detection, deep learning, fully convolutional Neural Network*.

I INTRODUCTION

The largest organ in the human body, the liver is situated on the right side of the stomach and is essential for metabolism, detoxification, and blood purification from the gastrointestinal tract. It also generates the proteins needed for blood coagulation. Primary or secondary liver cancer, including hepatocellular carcinoma (HCC) and intrahepatic cholangiomas, are serious health problems. Traditional diagnostic procedures include blood tests and biopsies, but CT scans and other imaging techniques are also being used more often [1].Deep learning (DL) algorithms have demonstrated potential in recent years for enhancing the segmentation and diagnosis of liver cancer. Methods like deep inception nets and probabilistic neural networks have proven to be highly accurate in identifying liver abnormalities.

For the purpose of segmenting livers and tumors, DL models, such as UNets and hybrid techniques like LeNet-5/ABC, have been created. These models handle issues including liver stretch and lesion complexity [2].Nevertheless, issues like erroneous border segmentation and under- or over-segmentation still exist. A number of methods, such as hybrid densely connected UNets and active contour models, attempt to solve these problems but have drawbacks. Models that use multi-scale context fusion networks and unique DL techniques integrated with Gaussian Mixture models for pancreatic cancer prediction are further developments [3].Furthermore, the necessity for invasive operations has decreased because to the developments in imaging tools like CT and MRI, which have made early identification and diagnosis easier. Deep learning techniques, in particular fully convolutional neural networks (FCNNs) and hybridized FCNNs (HFCNNs), have demonstrated great performance across a range of tasks, indicating its potential for liver cancer identification and segmentation [4]. All things considered, there is a great deal of promise for better patient outcomes through the identification and segmentation of liver cancer through the integration of deep learning algorithms with medical imaging. The resolution of outstanding issues and the improvement of automated liver cancer detection systems require more study and advancement in this field [5].

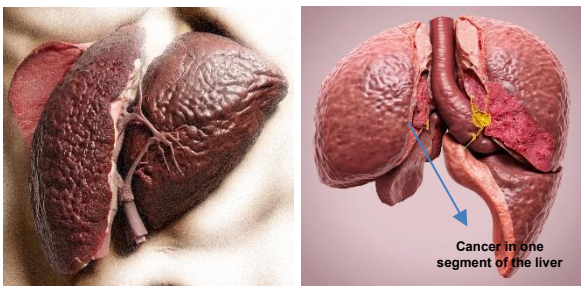

Fig1.Liver Tumors in Infancy

II LITERATURE REVIEW

The aforementioned studies advance the field of liver cancer detection and segmentation by putting forth novel ideas and methods. In an effort to improve detection accuracy, (MCG) method for segmenting liver tumors was first presented by

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Bai et al [1]. Offering a fresh viewpoint on the issue, Das et al. recommended using the Gaussian Mixture Model (WT-GMM) and Watershed Transform (WT-GMM) for identifying liver cancer[2]. With deep learning,According to Ben-Cohen et al., a fully convolutional network was used to identify liver lesions that produces better outcomes [4]. In order to analyze liver cancer using microarray data, Kim and Park developed the Hybrid Feature Selection (HFS) method, which combines neural network approaches with cross-validation [6]. Furthermore, the Hybridized Fully Convolutional Neural Network (HFCNN) for liver cancer was proposed in a paper. The aforementioned studies advance the field of liver cancer detection and segmentation by putting forth novel ideas and methods [7]. In an effort to improve detection accuracy, The Multi-scale candidate generation (MCG) method for segmenting liver tumors was first presented by Bai et al. Das et al.'s successful performance in the experimental results indicates that the Watershed Transform and Gaussian Mixture Model (WT-GMM) is a novel approach to the problem of liver cancer identification and segmentation [8]. Furthermore, another study created an ensemble segmentation technique based on deep learning to fill in the gaps in the literature about liver cancer detection and segmentation. [9].Together, these contributions improve the field by providing a variety of practical approaches to improve the precision and effectiveness of liver cancer detection and segmentation procedures [10].

RELATED WORK

The perfect accuracy score (93.5%) for A novel clinical range employing intelligent method is highly unusual and warrants further investigation to confirm its validity and generalizability [4]. Using a (HFCNN) built on a deep learning framework, liver cancer can be detected shows a promising direction with a reasonable accuracy of approximately 74.36%, highlighting the potential for mixed methodologies in improving prediction outcomes [1]. An overview of machine learning techniques used in gene expression research to predict cancer tested on the Indian Liver Patient Records dataset (13d167), SVC and MLP achieved the highest accuracy scores (73.68%), suggesting their suitability for this specific prediction task [5]. For transformer-based medical applications, unsupervised visual representation learning based on geometric pseudo-shape segmentation

tested accuracy score for the Indian Liver Patient Records dataset (72a5ee) suggests a different approach or model, achieving a higher accuracy of approximately 84.15%, indicating potential improvements or optimizations in the prediction process[7].

Table 1: Review & Analysis for Data Driven Approach of Liver Cancer Prediction

S. No	Dataset	Algorithm	Accuracy	Literature Survey	Source	Finding
1	The research uses CT scans to analyze a (HFCNN) model for liver tumor segmentation.	The algorithm proposed in the document is a (HFCNN) designed for liver tumor detection and segmentation in CT images.	a reasonable accuracy of approximately 74.36%.	studies various deep learning methods for liver cancer identification, including fully convolutional networks, hybrid feature selection strategies, Gaussian mixture models, multi-scale candidate generation, and Watershed Transform.	The Natural Science Foundation of Zhejiang Province funded this work with Grant LY17H160024. .	The suggested HFCNN model, which combines residual and pre-trained weights with feature extraction from Inception, performs admirably in the segmentation of liver cancers from CT images. With a notable improvement over current methods in terms of precision, accuracy, and dice similarity coefficient, the strategy intends to improve the accuracy of liver cancer diagnosis and segmentation. The study emphasizes how deep learning frameworks might improve the diagnosis and treatment planning of liver cancer.
2	A sample of 50,000 instances from seven different cancer patient types—liver, lung, kidney, breast, uterine, stomach, and intestine cancers—were used in the study.	(D-TSVR) technique that uses big data from the medical field to predict the recurrence of cancer.	It outperforms e-TSVR and Convolutional Neural Network (CNN) models with a prediction accuracy of over 91%.	The article addresses several approaches to cancer recurrence prediction, highlighting the value of applying machine learning algorithms and medical big data to increase prediction accuracy.	sponsored in part by the Hebei Province Natural Science Fund for Excellent Young Scholars, the National Natural Science Foundation of China, and the creative endeavors of Ph.D. candidates from Hebei .	An enhanced twin support vector regression (TSVR) algorithm was created and put to use. It used the cuckoo search algorithm for parameter optimization and dependent nearest neighbor (DNN) weighting for prediction accuracy. This improved TSVR algorithm showed a prediction accuracy for a variety of cancers that was higher than 91% for traditional convolutional neural networks and e-TSVR models.
3	The study involved 3411 participants for the screening of common cancers in China, using questionnaire data which contained 28 attributes.	The enclosed document covers the DBN-ELM-BP and PCA-ANN models. It highlights how, in terms of sensitivity, the DBN-ELM-BP model performs better than the PCA-ANN model.	The DBN-ELM-BP model outperformed the PCA-ANN model, with an accuracy of 98.03% on the training set and 98.24% on the test set	. analyzes machine learning techniques for cancer prediction with a focus on the Deep Belief Network-Extreme Learning Machine-Back Propagation (DBN-ELM-BP) and Principal Component Analysis-Artificial Neural Network (PCA-ANN) algorithms.	funded by Jinan Health Commission's Science and Technology Development Projects under Grant 2020-3-04.	With an accuracy of almost 98%, the DBN-ELM-BP model outperformed the PCA-ANN model in terms of both sensitivity and specificity, suggesting that it is a more cost-effective and non-invasive method of forecasting the likelihood of common malignancies.

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4	Utilized health examination data of 108 individuals, including 86 patients with nonalcoholic fatty liver disease (NAFLD) and 22 healthy controls.	outlines a unique clinical spectrum that can be used to diagnose nonalcoholic fatty liver disease (NAFLD) with intelligence.	With a 93.5% diagnostic accuracy rate.	Examines a range of intelligent computing methods for evaluating liver conditions, with a focus on Hidden Markov Models for NAFLD progression prediction.	Supported by Creative Commons Attribution 4.0 License and additional academic and research grants.	Created a novel clinical spectrum with non-invasive characteristics to evaluate NAFLD with Hidden Markov Models. shown effectiveness in differentiating between attribute credibility and determining stage development; has the potential to greatly improve NAFLD diagnosis and treatment planning.
5	The research looks at studies that use gene expression data from a range of cancer types, such as , liver ,breast, lung, kidney, ovarian, breast, central nervous system, and gallbladder, to identify biomarker genes and predict cancer.	Examining current cancer research that use gene expression data from different cancer types to predict survival, identify tumors, and stratify patients	SVC and MLP achieved the highest accuracy scores (73.68%).	Emphasizes the promise and difficulties of applying gene expression research to evaluate machine learning and deep learning techniques for cancer prediction.	Published under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License; affiliations: University of Babylon; University of Northampton.	Highlights the difficulties in analyzing gene expression data and the effectiveness of machine learning and deep learning in enhancing cancer prediction and classification, as well as the identification of biomarker genes. demonstrates the need for sophisticated computational techniques to improve forecast accuracy and the possible influence on tailored cancer treatment plans.
6	This investigation looked at 87 focal liver lesions (FLLs) in 72 patients who were at risk for hepatocellular carcinoma (HCC) using contrast-enhanced ultrasonography (CEUS).	Suggests a radiomics interpretable method for employing contrast-enhanced ultrasonography (CEUS) to distinguish benign from malignant focal liver lesions (FLLs).	87 FLLs from 72 individuals at risk for hepatocellular carcinoma were used in a clinical validation of the approach, which showed a balanced accuracy of 0.84 in differentiating between benign and malignant lesions	Explains the use of machine learning models to FLL characterization using CEUS, emphasizing the requirement for interpretable models that don't require motion compensation and only require a little amount of manual input.	supported, among others, by the 4TU Precision Medicine Program, the KNAW Van Leersum Grant, the Hanarth Fonds Fellowship for AI in Oncology, and the NIH/NCI.	The method achieves a balanced accuracy of 84% by combining spatiotemporal and textural analysis for FLL characterization. It emphasizes how crucial perfusion and microvascular architectural aspects are, along with the spatial features of contrast enhancement, for supporting FLL diagnosis.
7	The study examined CT records related to pancreatic cancer from 6,063 patients, liver cancer from 5,237 patients, and breast cancer from 127 patients using MRI technology.	It presents a Convolutional Pyramid Vision Transformer (CPT) that offers effective multi-scale feature capture and computational cost reduction.	Achieving a higher accuracy of approximately 84.15%.	Examines the application of unsupervised learning methods, with a special emphasis on segmenting geometric pseudo-shapes to enhance the learning of visual representations in medical imaging.	Funded by the Korea government (MSIT) Supported by the Korea Medical Device Development Fund .	Demonstrated that using geometric pseudo-shapes in an unsupervised learning context significantly enhances the performance of transformer-based models in classifying and segmenting medical images. The proposed Convolutional Pyramid vision Transformer (CPT) method, which incorporates unsupervised pretraining, showed superior classification and segmentation performance compared to existing methods.
8	The study uses the Liver Tumor Segmentation (LiTS) dataset, which consists of 200 liver CT images (130 for training and 70 for testing), to investigate the application of deep convolution neural networks (DCNN) for the diagnosis of liver cancer.	This paper presents Edge Strengthening Parallel UNet (ESP-UNet), a (DCNN) based automated liver cancer screening technique	The average Dice coefficient for segmentation was 0.92, and the accuracy of liver volume measurement was 97.22%.	Examines developments in the field of liver cancer detection, with a focus on deep learning techniques to enhance the precision of segmentation and classification.	The funding project for the Princess Nourah bint Abdulrahman University Researchers Supporting	introduced a lightweight sequential DCNN for liver cancer diagnosis and an Edge Strengthening Parallel UNet (ESP-UNet) model for liver segmentation. These models achieved good accuracy, precision, recall, and F1-score, indicating a significant improvement over previous approaches.

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9	Utilized data from 9,708 TCGA Whole-Exome Sequencing dataset covering 20 different cancer types.	a deep learning model that uses mutational signatures to predict cancer kinds and metastatic stages with a high degree of accuracy is introduced. DiaDeL differentiates between different types of cancer by examining the trinucleotide mutational pattern across different tumors and using non-negative matrix factorization (NMF) to uncover conspicuous mutational fingerprints	AUC of 0.88 and accuracies of 0.824 indicate its superior performance over numerous state-of-the-art approaches.	Discusses the application of mutational signatures in cancer genomics to the identification of tumorigenic pathways and the creation of strategies for precision medicine. It demonstrates several computational methods for examining mutational patterns in several cancer kinds.	With some funding from Taiwan's Ministry of Science and Technology (MOST)	a deep learning-based binary classifier that uses mutational signatures as input features to accurately identify different cancer types and predict metastatic stages. demonstrated better performance than six cutting-edge methods, with accuracy AUC metrics of 0.824 and 0.88, respectively.
10	Presents a sketch-supervised technique for tumor segmentation on histopathology photos, utilizing the public liver cancer dataset PAIP2019 .	A (HFCNN) is used by the framework to improve the detection accuracy of liver cancer. Precision in medical picture detection is the goal of the approach	The algorithm shows remarkable effectiveness in liver tumor identification with an average Dice coefficient of 0.92 for segmentation and an accuracy of 97.22% for liver volume measurement.	Focuses on merging CNN and Transformer networks for both local and global feature analysis in order to tackle the problem of accurate tumor segmentation with little operator input.	supported financially by the Chinese National Natural Science Foundation and the Zhejiang Provincial Natural Science Foundation.	Demonstrates the effectiveness of the proposed method in achieving high accuracy in tumor segmentation tasks, outperforming state-of-the-art methods on the datasets used.

Summary of these review works are as follows:

- To identify and segment lesions relevant to liver cancer, the publication uses a Hybridized Fully Convolutional Neural Network (HFCNN). This study examines the use and efficacy of this method in medical image processing to improve the accuracy of liver cancer detection. The HFCNN model, which uses deep learning to automate the detection process, has high accuracy in experiment results. This deep learning system can provide light on some aspects of the decision-making process through the analysis of inner layers and the description of attributes that contribute to predictions.
- This research describes a Deep-Twin Support Vector Regression (D-TSVR) method for predicting cancer recurrence using huge data from the medical profession. The D-TSVR algorithm shows promise in enhancing patient survival through targeted clinical therapies based on the recurrence time estimations. It is enhanced with a Dependent Nearest Neighbor (DNN) weighted technique and cuckoo algorithm for optimal parameter selection.
- The enclosed document contains a study on cancer prediction models. It highlights how, in terms of sensitivity and accuracy, the DBN-ELM-BP model performs better than the PCA-ANN model. These algorithms provide a non-invasive, cost-effective way to evaluate the probability of cancer based on data from screening questionnaires. The study demonstrates how early cancer screening and detection could be aided by these AI-powered models.
- This research report presents a novel clinical spectrum for the intelligent diagnosis of (NAFLD). It describes how to assess NAFLD phases using a Hidden Markov Model-based methodology in conjunction with the forward, Viterbi, and Baum-Welch algorithms. The study offers a non-invasive clinical spectrum with characteristics linked to different phases of non-alcoholic fatty liver disease, including biomarkers, symptoms, and risk factors. With a 93.5% diagnostic accuracy rate, the suggested approach has important implications for enhancing patient evaluations and assisting with medical decisions. The results of the study show that these creative techniques can advance medical knowledge and help doctors detect NAFLD more accurately and successfully.
- The paper "A Survey of Machine Learning Approaches Applied to Gene Expression Analysis for Cancer Prediction" provides a thorough overview of recent cancer studies using gene expression data from multiple cancer types (breast, gallbladder, lung, liver, kidney, central nervous system, and ovarian) for tumor identification, survival prediction, and stratification. It also highlights the use of machine learning techniques in identifying biomarker genes linked to specific cancer types, in addition to talking about the challenges and technical issues with the current cancer prediction models, particularly when comparing the levels of gene expression in cancerous and noncancerous tissues. The survey also examines the potential utility of identifying gene expression patterns for personalized treatment regimens and future cancer risk prediction.
- This study suggests using contrast-enhanced ultrasound (CEUS) to distinguish benign from malignant focal liver lesions (FLLs). The method is interpretable by radiologists. The technology offers a hybrid approach that, in order to eliminate manual input without requiring motion compensation, combines spatiotemporal analysis with texture analysis at critical points in the CEUS operation. Using 87 FLLs from 72 people who were at risk of hepatocellular carcinoma, a clinical validation of the method was carried out. The study emphasizes that to maximize diagnostic effectiveness, multiple extracted elements must be combined, including peak timing and textural properties at wash-in and peak enhancement.
- The study presents a novel approach to enhance transformer-based medical activities, particularly in the identification and categorization of cancers of the breast, pancreas, and liver. This approach combines geometric

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pseudo-shape segmentation with unsupervised visual representation learning. Convolutional Pyramid Vision Transformer (CPT) is presented, which provides efficient multi-scale feature capture and lowers computing costs. By integrating local spatial reduction with multi-kernel convolutional patch embedding, it is customized for medical imaging. The approach significantly outperformed state-of-the-art models using CT datasets for breast, pancreatic, and liver cancer, demonstrating its efficacy and promise for widespread application in medical diagnosis and treatment planning. In order to advance medical imaging analysis, the research emphasizes the significance of innovative neural network topologies and self-supervised learning.

- Edge Strengthening Parallel UNet (ESP-UNet), an automated liver cancer screening method based on Deep Convolutional Neural Networks (DCNNs), is presented in this research. Its goal is to reduce liver segmentation issues caused by under- and over-segmentation in CT images. The approach enhances liver CT picture texture, edge, and contrast by modifying double-stage Gaussian filtering. ESP-UNet, a novel segmentation technique, maximizes boundary area performance. This method showed notable improvements in accuracy when used in conjunction with a lightweight DCNN to represent features. The study demonstrates the superiority of ESP-UNet over traditional approaches, supporting more accurate liver tumor identification.
- A deep learning model is presented that use mutational signatures to accurately predict the types of cancer and the stages at which they will spread. By analyzing the trinucleotide mutational pattern in various tumors and employing non-negative matrix factorization (NMF) to find prominent mutational fingerprints, DiaDeL distinguishes between various cancer kinds. The model's ability to identify genes linked to cancer based on their salient features underscores its effectiveness and yields valuable insights for precision medicine and cancer treatment strategies.
- The publication presents an improved framework for tumor segmentation in histopathology images. The framework uses a (HFCNN) to increase the accuracy of liver cancer detection. The method uses numerous layers in the neural network to achieve precision in medical picture detection.In order to achieve precise segmentation, this method additionally adds a global normalized class activation map (CAM) to convolutional neural networks and transformers. The method's validation on medical imaging and diagnostics shows how promising it is.

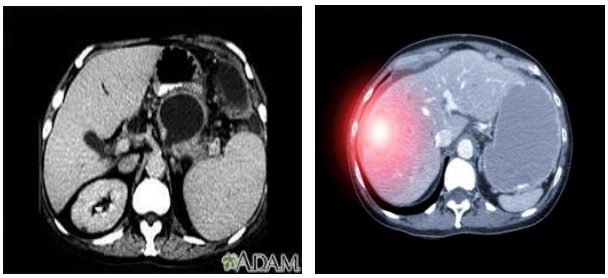

Fig2.Normal Abdominal CT scan & Liver Lesion Image [1].

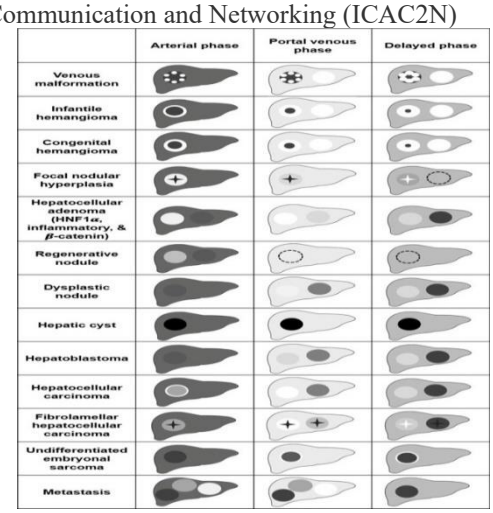

Fig.3. CT appearances of common liver lesions [23].

Grayscale and contrast-enhanced ultrasound images of common pediatric liver lesions in the early arterial, late arterial, portal venous, and washout phases were compared to the background liver parenchyma's improved appearance. Keep in mind that every circle depicts a possible location for the localized hepatic lesion. When a lesion appears more than once in the figure, it means that it could appear in different ways. Black or dark gray indicate non-enhancement, white indicates enhancement, and gray indicates some degree of enhancement. The stellate symbol and gray crosshairs represent the central scar. The symbol for iso-enhancement is a circle with a thin, dashed black line. White crosshairs indicate spoke-wheel centrifugal hyperenhancement of localized nodular hyperplasia.[23].

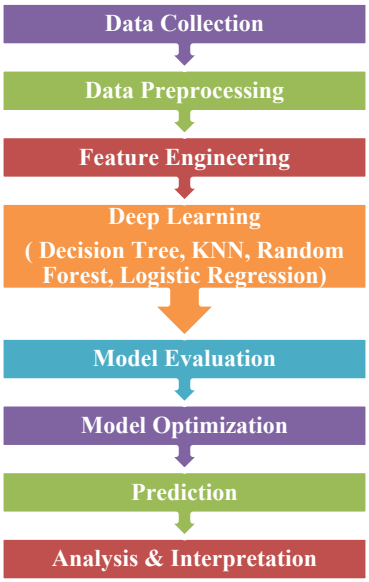

Fig4. Flowchart for liver cancer prediction

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Table 2: Indian Liver Patient Dataset Variable Description

Feature	Description(sample)
Age:	62
Gender:	female
TB (Total Bilirubin):	10.9
DB (Direct Bilirubin):	5.5
Alkphos (Alkaline Phosphatase):	699
Sgpt (Alamine Aminotransferase):	64
Sgot (Aspartate Aminotransferase):	100
TP (Total Proteins):	7.5
ALB (Albumin):	3.2
A/G Ratio (Albumin and Globulin Ratio):	0.74

Table 3: Statistical comparison for different approaches

Approach	F1 score	Recall	Precision	Accuracy
Decision Tree	0.33	0.37	0.31	0.63
KNN	0.85	0.95	0.76	0.74
Random Forest	0.82	0.87	0.78	0.72
Logistic Regression	0.26	0.17	0.62	0.76

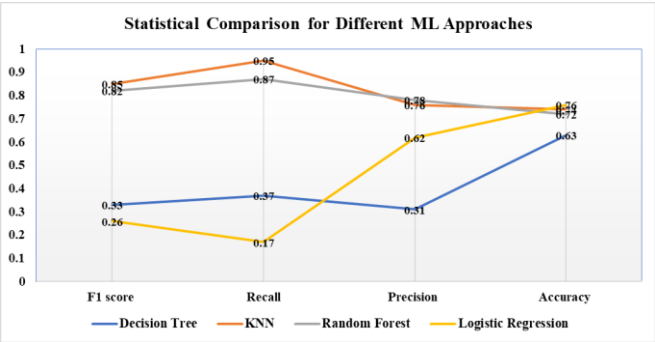

Fig 5: Statistical comparison for different ML Approaches

III. GAP ANALYSIS

Bridging the gap in radiological classification of different diseases is the main goal of the research study conducted in this context. The suggested (HFCNN) algorithm, with its excellent performance, signifies a noteworthy advance in the field of liver tumor segmentation. The study's Fully Convolutional Neural Networks (FCNNs) are noteworthy since they remove the requirement for explicitly defining particular radiological criteria in order to do picture recognition. This feature emphasizes how flexible and adaptable FCNNs are for analyzing medical images. The liver segmentation and metastases identification tasks have demonstrated the great efficacy and efficiency of fully convolutional architecture, indicating the promise of deep learning techniques to improve medical image processing. The progress that FCNNs have made in a number of medical applications highlights their adaptability and great potential.

IV. FUTURE SCOPE

To forecast the incidence, course, and reaction to therapy of liver cancer. Through the examination of large-scale databases containing patient characteristics, medical background, imaging information, and genetic markers, the research seeks to create a prediction model that can precisely predict the course of liver cancer. The aim of the project is to improve our knowledge of liver cancer and enable tailored healthcare interventions by means of feature selection, training, and evaluation. The predictive model that has been constructed may be used in clinical settings in the future, providing chances for early identification, risk assessment, and treatment planning.

V. CONCLUSION

The process of predicting the risk of developing liver cancer by machine learning entails evaluating patient data. For this, a variety of machine learning models each with advantages and disadvantages can be used.

Logistic Regression: This straightforward and easily comprehensible model is helpful in solving binary classification issues, such as the prediction of liver cancer. Because it offers probabilities of class membership, it makes it simple to comprehend the possibility that a patient has liver cancer.Although logistic regression works well for data that can be divided into linear segments, it may not be able to fully represent complicated, non-linear relationships in the data. It is, nevertheless, a useful starting point model for contrast.

Decision Tree: Decision trees are simple to understand and depict. Both numerical and categorical information can be handled by them, and they are able to extract non-linear correlations from the data. Overfitting of decision trees can occur, particularly with small datasets or when the tree grows too deep. Overfitting can be reduced with the use of pruning strategies and maximum depth settings.

Random Forest: Decision Tree ensembles used in Random Forests increase the robustness and accuracy of predictions. By averaging several trees, they lessen overfitting and can handle big datasets with intricate interactions. Performance-wise, Random Forests outperform individual Decision Trees in most cases. Although they are more accurate and resistant to overfitting than single Decision Trees, they are less interpretable.

KNN: KNN is a straightforward, non-parametric method that bases predictions on how similar two data points are to one another. Implementing and understanding it is simple. The distance measure and 'k' that are selected can have an impact on KNN. It can be computationally costly with huge datasets, but it might function effectively with enough data and appropriate feature scaling. Both irrelevant characteristics and noise can affect how well a KNN performs.

The ideal model for predicting liver cancer will vary depending on the dataset's unique properties and the desired ratio of performance to interpretability. Random forests are more accurate and robust than logistic regression, yet logistic

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regression is simpler and easier to understand. Decision trees provide a compromise between interpretability and non-linearity, whereas KNN offers simplicity but requires careful tweaking.

REFERENCES

1. Dong, X. *et al.* (2020) "Liver cancer detection using hybridized fully convolutional neural network based on deep learning framework," *IEEE access: practical innovations, open solutions*, 8, pp. 129889–129898. doi: 10.1109/access.2020.3006362.
2. Yang, A.-M. *et al.* (2021) "D-TSVR recurrence prediction driven by medical big data in cancer," *IEEE transactions on industrial informatics*, 17(5), pp. 3508–3517. doi: 10.1109/tii.2020.3011675.
3. Qi, H. *et al.* (2022) "Prediction methods of common cancers in China using PCA-ANN and DBN-ELM-BP," *IEEE access: practical innovations, open solutions*, 10, pp. 113397–113409. doi: 10.1109/access.2022.3215706.
4. Singh, A. *et al.* (2020) "A new clinical spectrum for the assessment of nonalcoholic fatty liver disease using intelligent methods," *IEEE access: practical innovations, open solutions*, 8, pp. 138470–138480. doi: 10.1109/access.2020.3011289.
5. Khalsan, M. *et al.* (2022) "A survey of machine learning approaches applied to gene expression analysis for cancer prediction," *IEEE access: practical innovations, open solutions*, 10, pp. 27522–27534. doi: 10.1109/access.2022.3146312.
6. Turco, S. *et al.* (2022) "Interpretable machine learning for characterization of focal liver lesions by contrast-enhanced ultrasound," *IEEE transactions on ultrasonics, ferroelectrics, and frequency control*, 69(5), pp. 1670–1681. doi: 10.1109/TUFFC.2022.3161719.
7. Viriyasaranon, T., Woo, S. M. and Choi, J.-H. (2023) "Unsupervised visual representation learning based on segmentation of geometric pseudo-shapes for transformer-based medical tasks," *IEEE journal of biomedical and health informatics*, 27(4), pp. 2003–2014. doi: 10.1109/JBHI.2023.3237596.
8. Dong, X. *et al.* (2020) "Liver cancer detection using hybridized fully convolutional neural network based on deep learning framework," *IEEE access: practical innovations, open solutions*, 8, pp. 129889–129898. doi: 10.1109/access.2020.3006362.
9. Napte, K. M., Mahajan, A. and Urooj, S. (2023) "Automatic liver cancer detection using deep convolution neural network," *IEEE access: practical innovations, open solutions*, 11, pp. 94852–94862. doi: 10.1109/access.2023.3307640.
10. Abdollahi, S., Lin, P.-C. and Chiang, J.-H. (2022) "DiaDeL: An accurate deep learning-based model with mutational signatures for predicting metastasis stage and cancer types," *IEEE/ACM transactions on computational biology and bioinformatics*, 19(3), pp. 1336–1343. doi: 10.1109/TCBB.2021.3115504.
11. Li, Y. *et al.* (2023) "Sketch-supervised histopathology tumour segmentation: Dual CNN-Transformer with global normalised CAM," *IEEE journal of biomedical and health informatics*, PP. doi: 10.1109/JBHI.2023.3289984.
12. J. He and B. Yu, *IEEE The 5th International Conference on Artificial Intelligence and Big Data*. China, 2022
13. Siróo and A. Almisreb, *IEEE 12th International Conference on Control System, Computing and Engineering (ICCSCE)*. Selangor, Malaysia; Penang, Malaysia, 2022
14. J. He and B. Yu, *IEEE The 5th International Conference on Artificial Intelligence and Big Data*. China, 2022.
15. K. Wang *et al.*, *2023 IEEE 3rd International Conference on Information Technology, Big Data and Artificial Intelligence*. Hong Kong, China.
16. M. Dr L Anand Dr, J. Dr, D. Dr, and A. Nagaraju, *Dr R G Vidhya Proceedings of the Fourth International Conference on Electronics and Sustainable Communication Systems (ICESC-2023) IEEE Xplore Part Number: CFP23V66-ART*.
17. D. Liu, J. Liao, and L. Liu, *Ubiquitous Intelligence & Computing, Advanced & Trusted Computing, Scalable Computing & Communications, Cloud & Big Data Computing, Internet of People and Smart City Innovation*. 2019.
18. Arshad, C. Dutta, T. Choudhury, and A. Thakral. (2018), "Liver Disease Detection Due to Excessive Alcoholism Using Data Mining Techniques." In IEEE International Conference on Advances in Computing and Communication Engineering (ICACCE), pp. 163-168.
19. Fernandez-Rozadilla C, Timofeeva M, Chen Z, Law P, Thomas M, Schmit S, Díez-Obrero V, Hsu L, Fernandez-Tajes J, Palles C, Sherwood K, Briggs S, Svinti V, Donnelly K, Farrington S, Blackmur J, Vaughan-Shaw P, Shu XO, Long J, Cai Q, Guo X, Lu Y, Brode.
20. N. Murphy *et al.*, "Body mass index and molecular subtypes of liver cancer," *J. Natl. Cancer Inst.*, vol. 115, no. 2, pp. 165–173, 2023.
21. Y. Takata *et al.*, "liver cancer risk: A pooled analysis of 12 prospective cohort studies," *J. Nutr.*, vol. 153, no. 7, pp. 2051–2060, 2023.
22. J. A. Lavery, P. C. Boutros, J. M. Scott, T. Tammela, C. S. Moskowitz, and L. W. Jones, "liver cancer analysis of postdiagnosis exercise and mortality," *J. Clin. Oncol.*, vol. 41, no. 32, pp. 4982–4992, 2023.
23. Schooler, G. R. et al. (2020) "Pediatric hepatoblastoma, hepatocellular carcinoma, and other hepatic neoplasms: Consensus imaging recommendations from American college of radiology pediatric liver reporting and data system (LI-RADS) working group," Radiology, 296(3), pp. 493–497. doi: 10.1148/radiol.2020200751.
24. Z. Deng *et al.*, "Data from lifetime body weight trajectories and risk : A large U.s. prospective cohort study,"2023.

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